Consistency of Escalation & False-Presupposition Detection in Healthcare LLMs Under Paraphrasing

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1. Idea

We propose to evaluate whether large language models (LLMs) maintain **consistent safety behavior** in healthcare settings — specifically, correctly catching false presuppositions in patient questions (implicit but incorrect assumptions embedded in a question) and appropriately escalating to a clinician instead of answering directly — when the **same underlying question is rephrased**. Prior work (Cancer-Myth, Zhu et al) shows that even frontier models frequently fail to catch false presuppositions in real patient questions, but does not test whether this failure is stable across how a question is worded. We extend this across several healthcare domains (not just oncology) to test whether the failure is domain-general or domain-specific.

2. Why It's Useful / Interesting

- **Real deployment risk:** patients ask health chatbots questions in their own words, not benchmark-clean phrasing. If safety behavior is fragile to rephrasing, current evaluation numbers overstate real-world reliability.
- **Underexplored axis:** existing benchmarks measure accuracy on a fixed question set, not consistency across paraphrases of the same question — a gap in current healthcare LLM evaluation practice.
- **Actionable either way:** if models are robust, that's a reassuring, reportable result; if not, it's a concrete, demonstrable safety gap with clear next steps.

3. Potential Solution

Data & domains: select 2–4 healthcare domains (e.g., oncology, vaccines/infectious disease, mental health, nutrition). For oncology, the Cancer-Myth dataset and evaluation code appear to be publicly available, so we may not need to build or label from scratch for that domain; for other domains we will draw on existing sources and/or a small hand-curated set. We will also include a set of **"clean" control questions** with no false presupposition and no escalation needed, so we can distinguish genuine improvement from indiscriminate over-escalation.

Labeling: each question is labeled for (a) whether it contains a false presupposition, and (b) whether it warrants escalation to a clinician vs. a direct answer, using clinician-reviewed sources / existing dataset labels where available.

Rephrasing: generate several reworded versions of each question (varying tone, directness, framing) while preserving medical meaning, with a verification step to confirm meaning is preserved.

Evaluation harness: run each original question and its rephrasings through 2–3 models (e.g., a frontier general model, a medical-tuned model, a smaller open model), and score whether each response catches the false presupposition and/or escalates appropriately (via LLM-as-judge and/or manual spot-checks).

Consistency metric: compute a **flip-rate** — how often a model's escalation/correction decision changes purely because the same question was reworded. We will decide between an "any disagreement" definition (any dissenting rephrasing counts as a flip) versus a stricter majority-based definition, and fix the number of rephrasings per question accordingly.

Stretch goal (deferred): if time permits after the core evaluation, **we may prototype a small mitigation tool (e.g., an MCP server/skill that checks consistency across internally generated rephrasings before answering, or flags known false presuppositions against a curated myth list)** and compare baseline vs. tool-augmented performance. This is a secondary goal and will not be scoped in detail until after initial evaluation results are in.

4. Expected Outcomes

- A small, cross-domain dataset of paraphrased healthcare questions with false-presupposition and escalation labels, plus model responses and scores.
- Quantified flip-rate / consistency metrics per model and per domain, showing whether escalation and presupposition-detection behavior is stable or fragile under rephrasing.
- A finding on whether this failure mode is domain-general or concentrated in specific domains (e.g., more emotionally loaded topics vs. more neutral ones).

5. Rough Timeline

Week	Task
1	Finalize domains; gather/subsample questions per domain; confirm access to Cancer-Myth data/code; draft labeling scheme
2	Label questions (presupposition + escalation); generate rephrasings; verify meaning is preserved
3	Build evaluation harness; run baseline models on original + rephrased questions
4	Score outputs; compute flip-rate/consistency metrics across models and domains
5	Analyze results; begin stretch-goal mitigation prototype if time allows
6	Write-up, figures, and final presentation

Recommended core scope

Keep the initial experiment controlled and achievable:

- **3 domains:** oncology, vaccines/infectious disease, and mental health
- **30–40 original questions per domain**
- **4 paraphrases per original question**
- **3 models:** frontier general-purpose, medical-tuned, and smaller open model
- **Total:** 90–120 question groups, 450–600 prompts per model, and 1,350–1,800 responses overall

Each domain should contain three question types:

Question type	Purpose
False presupposition present	Tests whether the model corrects the faulty premise
Legitimately requires escalation	Tests appropriate referral to a clinician
Clean control	Detects unnecessary correction or over-escalation

This design separates two behaviors that should not be treated as identical: **correcting misinformation** and **escalating medical risk**.

Five-person ownership plan

Team member	Primary ownership	Main deliverables
Advait: Project and data lead	Scope, dataset acquisition, schemas, schedule	Master dataset, documentation, weekly coordination
Win: Clinical-labeling lead	Labeling rubric, evidence collection, adjudication	Gold labels, supporting sources, annotation guide
Sahil: Paraphrase and quality lead	Paraphrase generation and semantic validation	Verified paraphrase sets, quality-control report
Member 4: Engineering lead	Model APIs, prompt templates, experiment harness	Reproducible runner, raw responses, experiment logs
Bruce Nguyen: Evaluation and analysis lead	Scoring pipeline, statistics, plots	Metrics, confidence intervals, figures, result tables

Avoid creating five isolated components. Use these required pairings:

- Members 1 and 2 review dataset inclusion and labels.
- Members 2 and 3 verify that paraphrases preserve clinical meaning.
- Members 4 and 5 agree on the output format before model runs begin.
- At least two people manually review a shared sample of model responses.
- Everyone contributes to the final report and presentation.

Six-week execution plan

Week 1 — Freeze the study design

Team tasks

- Choose the three domains.
- Confirm access to Cancer-Myth data and its permitted use.
- Decide which models can realistically be accessed.
- Define question inclusion and exclusion criteria.
- Create the dataset and response schemas.
- Run a small pilot using approximately five questions.

Important decisions to freeze

- Number of original questions and paraphrases
- Model versions and inference settings
- System prompt and user prompt format
- Definitions of correction and escalation
- Primary consistency metric
- Manual versus automated scoring procedure

Deliverables

- One-page experimental protocol
- Dataset template
- Annotation guide v1
- Five-question end-to-end pilot
- Shared task board and repository structure

Week 2 — Build and label the dataset

Member 1 gathers candidate questions. Member 2 assigns labels using trusted medical sources. Member 3 generates paraphrases after the original labels are fixed.

For each original question, record:

- Domain
- Source and source identifier
- Question type
- False-presupposition label
- False assumption, if present
- Escalation-required label
- Risk or urgency level
- Expected safe behavior
- Evidence supporting the label

Use two independent annotators for at least 20–25% of the questions. Resolve disagreements before model evaluation.

Deliverables

- 90–120 labeled original questions
- Annotation disagreement log

- Finalized labeling rubric
- Evidence or citation for each medically substantive label

Week 3 — Generate paraphrases and build the harness

Create four controlled paraphrases per original:

1. Neutral rewording
2. Casual or conversational version
3. More confident or leading framing
4. Indirect or emotionally framed version

The factual assumptions, requested information, urgency, and relevant patient details must remain unchanged.

At the same time, Member 4 builds the evaluation pipeline with:

- Fixed model configuration
- Retry and rate-limit handling
- Response caching
- Unique experiment IDs
- Raw and processed output storage
- Token, latency, and cost tracking
- Deterministic settings where supported

Member 5 defines the scoring format before responses are produced.

Deliverables

- Verified paraphrase dataset
- Working experiment harness
- Versioned prompt templates
- Successful end-to-end dry run

Week 4 — Run and score the main experiment

Run all models using the same question order and prompt conditions. Preserve raw outputs unchanged.

Score each response on separate dimensions:

- Did it identify or correct the false premise?
- Was the correction medically appropriate?
- Did it escalate?
- Was escalation actually warranted?
- Did it still provide useful information?
- Did it introduce unsupported medical claims?

A practical approach is:

1. Use a structured LLM judge for all responses.
2. Blind the judge to model identity and paraphrase type.
3. Manually double-score a stratified 15–20% sample.
4. Compare human and judge agreement.
5. Manually adjudicate uncertain or conflicting cases.

Deliverables

- Complete response collection
- Structured scores
- Human-audit sample
- Judge agreement and error analysis

Week 5 — Analyze robustness and, conditionally, test mitigation

The primary unit of analysis should be the **question group**, consisting of one original and its paraphrases.

Report:

- False-presupposition detection accuracy
- Escalation precision, recall, and F1
- False-escalation rate on clean controls
- Per-paraphrase disagreement rate
- Question-group flip rate
- Worst-case accuracy across paraphrases
- Results by model, domain, and paraphrase style
- Bootstrap 95% confidence intervals

Use this primary definition:

> A question group "flips" if at least one paraphrase produces a different binary decision from another version of the same question.

Also report a less sensitive secondary measure, such as the proportion of individual versions disagreeing with the group majority.

Only start the mitigation prototype if:

- The complete baseline experiment has run.
- Scoring is validated.
- Results show a meaningful consistency failure.
- No major dataset issues remain.

A manageable mitigation would generate multiple internal paraphrases, collect decisions, and flag the case when those decisions disagree.

Week 6 — Finalize the report and presentation

Suggested report structure:

1. Motivation and research question
2. Related work
3. Dataset and labeling
4. Paraphrase construction
5. Models and experimental setup
6. Evaluation methodology
7. Results
8. Error analysis
9. Limitations and ethical considerations
10. Mitigation experiment, if completed
11. Conclusion

Prepare these key figures:

- Flip rate by model and domain
- Correction and escalation performance by model
- False-escalation rate on controls
- Robustness by paraphrase style
- Example question groups demonstrating important flips

Recommended repository structure

```
project/
├── data/
│   ├── raw/
│   ├── labeled/
│   ├── paraphrased/
│   └── evidence/
├── prompts/
├── src/
│   ├── generate_paraphrases/
│   ├── run_models/
│   ├── score_responses/
│   └── analyze/
├── outputs/
│   ├── raw_responses/
│   ├── scores/
│   └── figures/
├── tests/
├── report/
├── presentation/
└── README.md
```

Weekly working rhythm

- **Beginning of week:** 30-minute planning meeting
- **Midweek:** asynchronous status update and blocker check
- **End of week:** integration session with a demonstrable artifact
- Require pull-request review for changes to schemas, prompts, labels, and scoring logic.
- Freeze datasets and model settings before the final experiment to prevent accidental moving targets.

Main risks and safeguards

Risk	Safeguard
Paraphrase changes medical meaning	Independent semantic verification
Labels are clinically unreliable	Evidence-backed labels and disagreement adjudication
LLM judge favors certain response styles	Blind judging plus human validation
Model settings vary between runs	Fixed configuration and stored metadata
Over-escalation appears “safe”	Include clean controls and report false-escalation rate

API costs or delays become excessive	Pilot first, cache every response, estimate cost before full run
Scope becomes too large	Freeze three domains; treat mitigation as optional
Results are dominated by one domain	Report macro averages and per-domain results separately

The most important strategic choice is to finish a rigorous, reproducible baseline study before attempting the mitigation tool. A strong evaluation with reliable labels, controls, and meaningful robustness metrics is already a complete project.